

LabGuardian: Geometry-Constrained Pin Pose Estimation for Visual Breadboard Reconstruction

1st Su'an Liu*

School of Electronic Engineering
Beijing University of Posts
and Telecommunications
Beijing, China
buptliusuan@gmail.com

2nd Xinran Zhang

School of Electronic Engineering
Beijing University of Posts
and Telecommunications
Beijing, China

3rd Jiali Ruan

School of Electronic Engineering
Beijing University of Posts
and Telecommunications
Beijing, China

Abstract—Recovering the wiring of a breadboard circuit from a single photograph is a fine-grained visual structure problem: holes are separated by a 2.54 mm pitch, component terminals are thin and frequently occluded, and a displacement of one grid cell changes the inferred electrical connection. Component detectors stop at bounding boxes and cannot say which hole a lead enters, whereas end-to-end multimodal models may describe a plausible topology that the image does not support. We present LabGuardian, a vision pipeline that formulates terminal localization as component-conditioned pose estimation. A full-frame keypoint model predicts component instances together with ordered terminal keypoints, a planar homography rectifies the board to a canonical metric plane, and a geometry-constrained snap-to-grid stage maps each rectified terminal to a discrete hole under package and board feasibility constraints while retaining ranked alternatives for ambiguous terminals. Every downstream relation stays linked to its originating instance, keypoint and candidate hole, so the reconstruction is auditable rather than asserted. On a disjoint validation split, component detection reaches 0.991 mAP50 and 0.786 mAP50-95 and pin pose estimation reaches 0.947 and 0.829. INT8 execution on an Intel Core Ultra 5 225U neural processing unit runs at 13.37 ms mean latency, 15.61 ms at the 99th percentile and 74.7 images per second within 8.53 W package power.

Index Terms—pose estimation, keypoint localization, homography, visual circuit reconstruction, edge inference

I. INTRODUCTION

Converting a photograph of a physical circuit into an explicit structural description is a prerequisite for automated inspection of hand-wired hardware, for robotic assembly verification, and for assistive systems that check a prototype against an intended design. Breadboards make this problem concrete and unusually strict. Holes are arranged on a regular 2.54 mm lattice, and the electrical meaning of a component is determined entirely by which holes its terminals occupy. Two reconstructions that differ by a single grid cell describe different circuits: one may be the intended amplifier, the other an open loop or an unintended short. Accuracy therefore has to be defined at the level of individual holes rather than of component bodies.

Most visual pipelines stop before that level. A detector that returns a box labeled *resistor* identifies the presence of a device but not where its leads enter the board, and cropping the

box before regressing terminals removes exactly the evidence needed: axial leads extend well beyond the component body, and the surrounding lattice provides the reference against which a terminal position becomes meaningful. The natural representation is an instance-conditioned set of terminal keypoints predicted in the full image.

A second difficulty is geometric rather than semantic. Nearest-neighbor matching in image coordinates is not invariant to viewpoint: under an oblique camera the row spacing compresses, and a terminal can be attracted to the wrong row even when its pixel localization is correct. Purely learned image-to-topology models, in turn, have no explicit mechanism to enforce the planar lattice, the separation across the central trench, the span of an axial device, or the pin order of a dual in-line package. Multimodal language models provide flexible visual semantics [1], but a free-form description cannot guarantee that a claimed connection corresponds to observed pixels; we refer to such unsupported structural statements as *topology hallucination*. In a verification setting this failure mode is worse than a missed detection, because an unsupported claim is indistinguishable from a supported one.

LabGuardian therefore treats reconstruction as a sequence of visual estimation and geometric projection steps, each of which produces an inspectable intermediate result. A full-frame pose network predicts component instances and ordered terminal keypoints; a planar homography maps the board to a canonical metric plane; and a geometry-constrained snap-to-grid stage converts rectified coordinates into discrete hole hypotheses, keeping ranked alternatives wherever the observation does not support a unique assignment. Electrical interpretation and symbolic comparison consume this pin-hole graph but never modify it, so a structural conclusion can always be traced back to a bounding box, a keypoint, a rectified coordinate and a candidate hole.

The contributions of this paper are as follows.

- We formulate terminal localization as component-conditioned pose estimation on the full board image, and associate pin hypotheses with component instances through a multi-cue, category-gated matching score with explicit abstention.

*Corresponding author.

- We introduce a projective-to-discrete geometric interface that couples homography-based rectification with a feasibility-constrained snap-to-grid assignment, and that represents residual hole-level ambiguity explicitly instead of resolving it silently.
- We report accuracy and heterogeneous edge inference for the complete visual stage: 0.947 pin mAP50 and 0.829 pin mAP50-95, and 13.37 ms mean latency at 74.7 images per second within 8.53 W on an integrated neural processing unit.

II. RELATED WORK

A. Keypoint and Pose Estimation

Pose estimation maps an image to structured landmarks instead of a category or a box. High-resolution architectures such as HRNet improve spatial precision by maintaining multi-scale feature streams throughout the network [2], while single-stage formulations such as YOLO-Pose regress boxes and keypoints jointly and retain instance-level grouping in one forward pass [3], [4]. We adopt this heatmap-free family but reinterpret the pose: the instance is a rigid electronic component and the landmarks are its electrical terminals. Unlike human joints, these landmarks are small, often collinear, and constrained both by package geometry and by the regular lattice that receives them, which makes the surrounding context at least as informative as the appearance of the instance itself.

B. Planar Rectification and Metric Grids

Projective rectification of a plane from four correspondences is a standard tool for normalizing viewpoint [5], [6]. A breadboard is a favorable case, since its working surface is approximately planar and carries a repeated metric grid that provides landmarks. Rectification alone, however, does not decide between two adjacent holes when a terminal projects near their common boundary. Our contribution at this stage is not the homography itself but its coupling to a discrete assignment layer that encodes board topology, package type, terminal span, collinearity and trench separation, and that reports the constraint responsible for each rejection.

C. Circuit Understanding and Symbolic Verification

Work on circuit data has largely addressed schematic-level recognition, printed circuit board inspection, or already symbolic netlists [7]. Simulation and graph matching operate on a symbolic circuit that is assumed to be available [8], [9], and language-model approaches to electronic design consume textual or structural descriptions rather than raw photographs [10]. Conversely, generic multimodal models summarize visual content without providing a deterministic pixel-to-hole correspondence. The interface between these regimes, namely the recovery of terminals and their canonical board locations from a photograph, is the gap addressed here. Symbolic comparison is retained in our system only as a downstream consumer of the visual reconstruction and is not claimed as a contribution.

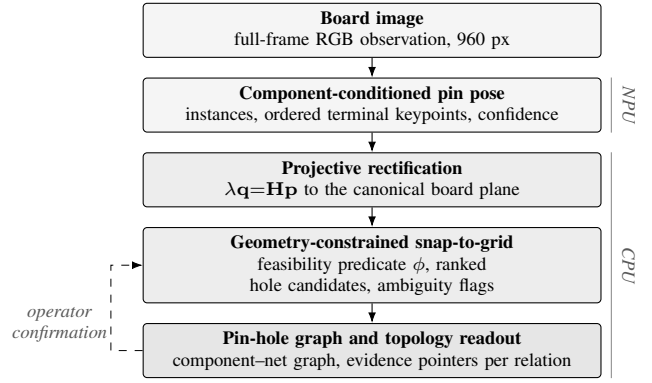


Fig. 1. Visual reconstruction pipeline. Perception, rectification and discrete assignment are separated so that every stage can be inspected independently. Topology readout consumes the pin-hole graph but does not alter the visual observations; a confirmed hole assignment re-runs only the geometric and symbolic stages. Vertical rules on the right indicate the execution unit that hosts each stage on the target platform.

III. METHOD

A. Problem Formulation

Let an RGB image $I \in \mathbb{R}^{H \times W \times 3}$ show component instances $\mathcal{C} = \{C_i\}$. Each instance carries a class t_i , a bounding box b_i , and an ordered set of terminal keypoints $K_i = \{\mathbf{p}_{ik}\}$ with visibility flags. A board model supplies canonical hole centers $\mathcal{H} = \{\mathbf{c}_h\}$ together with feasibility relations induced by the board layout and by component packaging. The task is to estimate

$$F : I \longrightarrow \{(C_i, k, h, z_{ik})\}, \quad (1)$$

where h is the assigned hole and z_{ik} records keypoint confidence, rectified projection distance, candidate rank, visibility and an ambiguity flag. The formulation deliberately precedes electrical interpretation: it states what the image supports, not whether the circuit is correct. Fig. 1 summarizes the resulting stages.

B. Component-Conditioned Pin Pose Estimation

The pose network processes the complete image at 960 px resolution and, for every detected component, regresses a box, a class score, terminal coordinates, per-keypoint visibility and per-keypoint confidence. Full-frame inference is a deliberate choice: it preserves leads that leave the component body and the lattice that surrounds them, both of which a crop would discard. Seven classes are annotated directly: resistor, ceramic capacitor, electrolytic capacitor, diode, light-emitting diode, jumper wire and three-terminal transistor. Dense dual in-line packages are treated differently. Their pins occupy only a few pixels each and direct keypoint regression is unstable, so pin positions are derived from rigid package geometry once the package and its orientation have been detected.

Component and pose hypotheses are produced by two heads and must be associated before a netlist can be built; a terminal claimed by a neighboring device propagates into a wrong

electrical relation. For component i and pose hypothesis j we compute

$$S_{ij} = w_1 \text{IoU}(b_i, b_j) + w_2 \exp\left(-\frac{\|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|_2^2}{2\sigma_i^2}\right) + w_3 \rho_{ij} + w_4 \gamma_{ij} + w_5 c_j, \quad (2)$$

where $\boldsymbol{\mu}$ denotes a box center, ρ_{ij} is the fraction of keypoints falling inside a dilated component box, γ_{ij} measures the consistency between keypoint span and component extent, and c_j is the detection confidence. Pairs with incompatible classes are removed, the remaining pairs are sorted by S_{ij} and assigned greedily one to one, so a terminal cannot be owned twice. If the best score of a pose hypothesis falls below an acceptance threshold, the hypothesis is left unmatched and flagged rather than attached to the closest device. Abstention is preferred here because an unmatched terminal is visible to an operator, whereas a wrong owner silently corrupts the netlist.

C. Projective Rectification

Let $\mathbf{p} = [x_p, y_p, 1]^\top$ be a terminal keypoint in image coordinates. Four board landmarks determine a nonsingular planar homography $\mathbf{H} \in \mathbb{R}^{3 \times 3}$, and the canonical coordinate $\mathbf{q} = [u, v, 1]^\top$ follows from

$$\lambda \mathbf{q} = \mathbf{H} \mathbf{p}, \quad u = \frac{h_{11}x_p + h_{12}y_p + h_{13}}{h_{31}x_p + h_{32}y_p + h_{33}}, \quad v = \frac{h_{21}x_p + h_{22}y_p + h_{23}}{h_{31}x_p + h_{32}y_p + h_{33}}, \quad (3)$$

with $\lambda \neq 0$ the homogeneous scale and h_{ij} the entries of $\mathbf{H}$. Equation (3) removes projective distortion before any discrete decision is taken, which makes the acceptance radius of the next stage a physical quantity instead of a viewpoint-dependent one. When landmark calibration fails, a synthetic lattice is used as a fallback, but every assignment produced under the fallback is marked so that it is not treated as equally reliable evidence.

D. Geometry-Constrained Snap-to-Grid

Given the rectified coordinate and the canonical hole set, the candidate assignment and its acceptance rule are

$$h^* = \arg \min_{h \in \mathcal{H}} \|\mathbf{q}_{1:2} - \mathbf{c}_h\|_2, \quad a = \frac{d_{h^*}}{s_g}, \quad (4)$$

$$\text{accept} = (a \leq \tau) \wedge (\phi(h^*, t_p, K_i) = 1), \quad (5)$$

where d_{h^*} is the rectified snapping distance, s_g the physical grid pitch, τ a normalized threshold, t_p the terminal or package type, and ϕ a feasibility predicate. The predicate rejects assignments that leave the board region, cross the central trench where the two halves are electrically separate, violate the span of an axial device, break the collinearity of a three-terminal device, or contradict package-specific pin order. Normalizing the snapping distance by s_g in (4) makes τ independent of output resolution.

Rejection does not discard the observation. The mapper records the ranked candidate holes, their normalized distances,

the violated constraint and a confidence flag, so a terminal that projects near the boundary between two holes is reported as ambiguous rather than resolved by an arbitrary tie-break (Fig. 3). This is a deliberate design decision: dense millimeter-scale holes are an intrinsically hard localization target, a residual hole-level error cannot be driven to zero by a stronger backbone alone, and a sub-pixel residual that is silently rounded becomes a false structural claim. Where a low-confidence assignment matters, a single operator confirmation resolves it and only the geometric and symbolic stages are recomputed; the perception result is not discarded.

E. Topology Readout

Accepted pin-hole relations are grouped into electrical nodes using the known breadboard layout, in which each five-hole strip is internally connected and the power rails span the board. A disjoint-set structure merges the strips joined by detected jumper wires, and the result is a component-net graph in which nets are nodes and non-wire components are edges. Physical holes remain the only source of node identity, so a corrected hole assignment regenerates node labels instead of inheriting stale ones.

The graph can be exported as a topology-level netlist and compared against a reference by subgraph isomorphism [9], which yields structured mismatch codes. These operations add no visual evidence and are not evaluated as part of our claims; their role is to test whether the reconstructed structure is usable downstream. Because each mismatch retains pointers to the originating instance, keypoint, rectified coordinate and candidate hole, a conclusion can be audited back to the pixels that produced it.

IV. EXPERIMENTS

A. Implementation and Data

Experiments use a private dataset of real top-view breadboard photographs covering six textbook analog circuit topologies: a first-order RC network, a common-emitter amplifier, a differential pair, and inverting, integrating and summing amplifiers built around a general-purpose operational amplifier. The images were captured under varying viewpoint and illumination and include wire overlap and partial occlusion, so no synthetic augmentation of the scene layout was required. Annotations combine component boxes with up to three ordered terminal keypoints per instance and per-keypoint visibility. Training, validation and test splits are disjoint, and the test split is excluded from optimization and model selection.

The pose model is YOLOv8s-pose [4], trained for 100 epochs at 960 px input with batch size 8, standard color, translation and scale augmentation, and mosaic augmentation disabled for the final ten epochs. Post-training INT8 quantization used 144 calibration images and reduced the exported model from 22 MB to 11 MB. All deployment measurements were taken on an edge platform built around an Intel Core Ultra 5 225U with 8 GB of memory, which exposes CPU, integrated GPU and neural processing unit execution paths through a common runtime [11], [12]. Between 60 and 1153

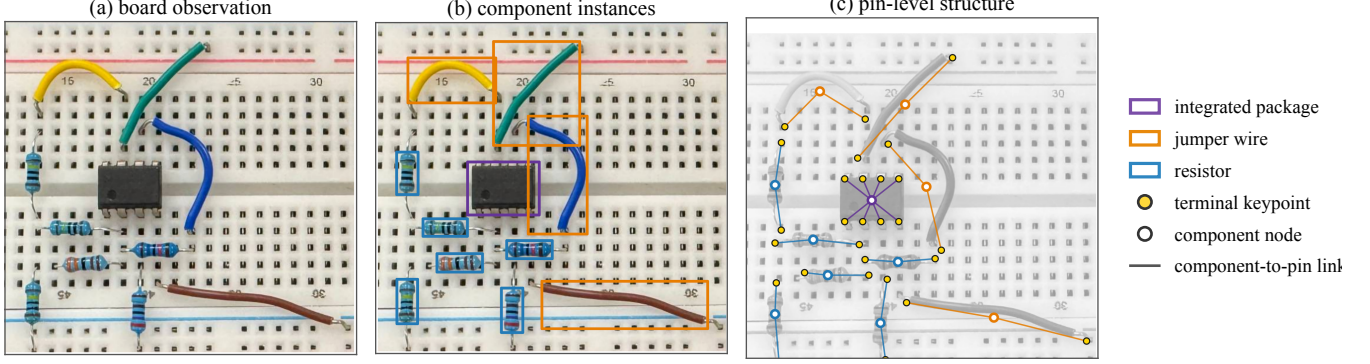


Fig. 2. Reconstruction on a real board. (a) full-frame observation; (b) component instances with class-coded boxes; (c) the instance-linked pin representation after geometric mapping, where each terminal keypoint is bound to a component node and to the hole it occupies. Panel (c) is the interface consumed by topology readout; it adds no visual evidence of its own.

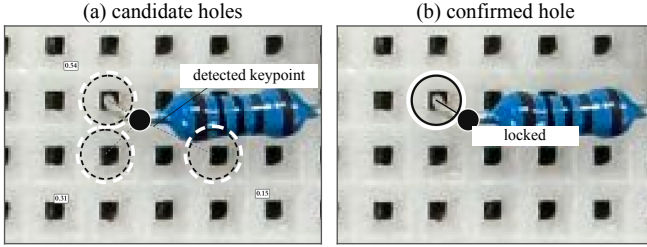


Fig. 3. Hole-level ambiguity. (a) A terminal that projects between adjacent holes is retained as several ranked candidates with their scores instead of being forced onto the nearest one. (b) After confirmation the terminal is bound to a single hole and the downstream graph is recomputed without re-running perception.

serial inferences were collected per configuration, and package power was sampled from the running average power limit interface at 0.25 s intervals.

B. Detection and Pin Pose Accuracy

Table I reports validation accuracy after 100 epochs. The keypoint head keeps precision and recall above 0.95 despite the small spatial support of a terminal, which indicates that terminals of the annotated classes are reliably found and attributed to the correct instance. The gap between pin mAP50 and mAP50-95 is the more informative number for our purpose: coarse keypoint detection is largely solved, while strict localization is not, and it is precisely the strict regime that determines whether a terminal falls on the correct side of a hole boundary. This is the empirical motivation for treating rectification and constrained assignment as separate, inspectable stages rather than expecting the pose head to resolve hole identity by itself.

TABLE I
VALIDATION ACCURACY AFTER 100 EPOCHS

Prediction target	Precision	Recall	mAP50	mAP50-95
Component boxes	0.991	0.989	0.991	0.786
Pin keypoints	0.955	0.954	0.947	0.829

TABLE II
LATENCY AND THROUGHPUT OF THE POSE MODEL ACROSS EXECUTION UNITS

Device	Prec.	Load (s)	Mean (ms)	P95 (ms)	P99 (ms)	img/s
CPU	FP16	0.14	92.44	96.74	98.92	10.8
CPU	INT8	0.24	29.02	30.13	30.65	34.5
iGPU	FP16	0.43	26.87	27.32	28.18	37.2
iGPU	INT8	2.85	18.26	19.29	20.11	54.7
NPU	FP16	1.01	16.55	16.64	17.67	60.4
NPU	INT8	1.20	13.37	13.75	15.61	74.7

C. Heterogeneous Edge Inference

Table II lists latency and throughput for the six device-precision configurations. INT8 execution on the neural processing unit attains the lowest mean and tail latency, reducing mean latency by 85.5% relative to CPU FP16 and raising throughput from 60.4 to 74.7 images per second relative to FP16 on the same unit. The 15.61 ms 99th-percentile latency is the relevant figure for interactive use, since it bounds the delay of the visual stage rather than its average cost.

Fig. 4 shows the package power trace of the three INT8 configurations measured in one session. Two observations follow. First, the neural processing unit sustains the workload at 8.53 W package power, roughly 4.1 W above the 4.4 W idle baseline. This corresponds to 114.2 mJ per inference, or 18.1 images per second per watt of incremental power, whereas the CPU configuration draws 26.4 W at 34.5 images per second and therefore spends roughly seven times more energy on

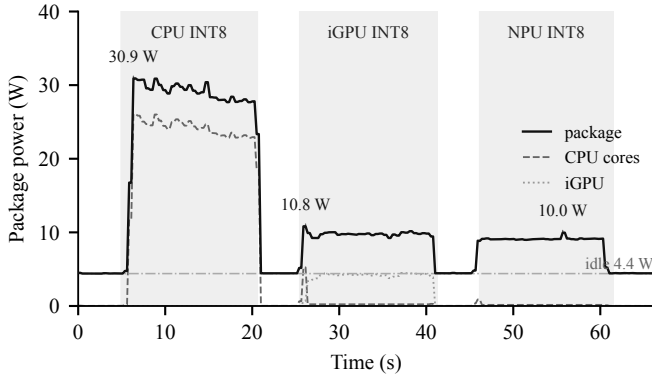


Fig. 4. Package power during three INT8 inference phases on the same platform. Peak package power is annotated per phase. While the neural processing unit is loaded, CPU-core and integrated-GPU power remain at the idle baseline.

the same inference. Second, during neural processing unit execution the CPU-core and integrated-GPU power curves stay near their idle level, so the units are genuinely free for other work. This is what makes the division of labor in Fig. 1 practical on one 8 GB platform: perception is resident on the neural processing unit while graph construction, rectification bookkeeping and verification run on the CPU.

D. Qualitative Analysis

Fig. 2 shows a complete reconstruction. Boxes and classes are recovered on a cluttered board, and the pin-level panel makes explicit which terminal belongs to which instance and which hole each terminal occupies. Two failure sources are visible in this representation and are worth separating. Component boxes can be correct while terminal endpoints are partially hidden by jumper wires, and full-frame inference is what preserves the visible lead direction and neighboring lattice needed to place such endpoints. Independently, a terminal can be localized accurately and still fall near a boundary between two holes after rectification, as in Fig. 3; this is a geometric condition, not a perception error, and the pipeline reports it as such.

Fig. 5 shows the structure obtained from the same board after grouping: 11 components and 28 terminals resolve into 10 electrical nodes, with power and ground rails distinguishable from signal nodes. The example is included as evidence that the visual output is complete enough to be consumed symbolically, not as a claim about diagnostic accuracy. On this platform the deterministic path from a rectified pose result to a topology comparison completes in under 100 ms, so the visual stage dominates the end-to-end cost.

E. Cost of the Auditable Interface

Keeping perception, geometry and interpretation separate has an engineering cost, which we report for completeness. An optional on-device explanation module, used to verbalize structured findings under the fixed evidence produced by the symbolic stage, occupies 941.5 MB after INT4 weight compression, down from 3.1 GB, and peaks at 1.36 GB resident

memory with 63.9 ms to the first token and 23.2 tokens per second on the integrated GPU [13]–[15]. It therefore coexists with the vision model inside the 8 GB budget. Because this module only rephrases findings it is given and cannot alter the reconstruction, it is outside the scope of the visual claims evaluated above.

V. DISCUSSION AND LIMITATIONS

The results above quantify the perception stage and the cost of deploying it: terminals of the annotated classes are localized and attributed accurately, and the visual stage runs at interactive rates within an embedded power budget. The geometric stage is evaluated qualitatively here, and a quantitative hole-level benchmark remains open. Reporting pin-to-hole assignment accuracy, or an ablation that separates projective rectification from unconstrained nearest-neighbor snapping, requires per-terminal hole labels that the present annotation does not carry, and we prefer to state the protocol rather than approximate it from pose average precision: normalized keypoint error and PCK for the continuous stage, hole-assignment accuracy after rectification and snapping for the discrete stage, ambiguity-rejection precision over the terminals rejected by ϕ , each stratified by viewpoint obliquity and occlusion level. Collecting these labels for a held-out subset is the next step, and it is what would isolate the contribution of the geometric layer from that of the pose model.

Two further limitations bound the present results. The dataset is private and its keypoints denote rigid terminals rather than articulated joints, so results are not directly comparable with public pose benchmarks; releasing the annotations and the board geometry would improve reproducibility more than any additional architectural change. And the symbolic stage can detect that a reconstruction is structurally inconsistent with a reference, but it cannot certify that an individual keypoint or hole assignment was correct. Visual and structural metrics must therefore continue to be reported separately, which is also why the system exposes ambiguity to an operator instead of resolving it internally.

VI. CONCLUSION

We presented LabGuardian, which reframes breadboard understanding as pin-level pose estimation coupled to an explicit geometric interface. Full-frame, component-conditioned keypoint prediction preserves the lead and lattice context that cropping destroys; a planar homography maps arbitrary viewpoints to a canonical board; and a feasibility-constrained snap-to-grid stage converts rectified coordinates into discrete hole hypotheses while keeping residual ambiguity visible. The reconstruction is audible because every downstream relation remains bound to the image evidence that produced it. Validation accuracy and cross-device measurements show that the approach is spatially accurate at the pose level and suitable for real-time execution on an integrated neural processing unit. The immediate next step is the controlled geometric ablation and the pin-to-hole benchmark described above, which are

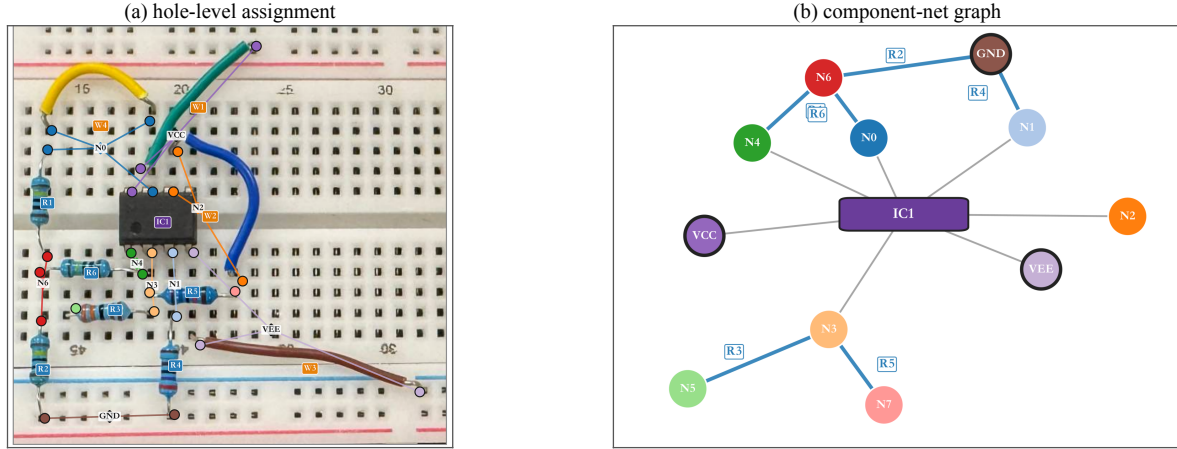


Fig. 5. Topology readout for the board of Fig. 2: 11 components, 28 terminals and 10 electrical nodes. (a) Terminals bound to holes, with one color per electrical node and jumper wires merged into their nodes. (b) The resulting component-net graph, in which nets are nodes and non-wire components are edges. Node identity comes from physical holes, so a corrected assignment regenerates the graph.

required to quantify the geometric stage independently of the pose model.

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